Question-1

Problem Statement – Madhu went to a movie with his friends in a theatre and during break time he bought pizzas, puffs and cool drinks. Consider the following prices:

- Rs.100/pizza
- Rs.20/puffs
- Rs.10/cooldrink

Generate a bill for What Madhu has bought.

Sample Input 1:

- Enter the no of pizzas bought:10
- Enter the no of puffs bought:12
- Enter the no of cool drinks bought:5

Sample Output 1:

Bill Details

- No of pizzas:10
- No of puffs:12
- No of cooldrinks:5
- Total price=1290

ENJOY THE SHOW!!!

Question-2

Problem Statement – Bumra wants a magic board, which displays a character for a corresponding number for his science project. Help him to develop such an application.

For example when the digits 65,66,67,68 are entered, the alphabet ABCD are to be displayed.

[Assume the number of inputs should be always 4]

Sample Input 1:

- Enter the digits:

  65
  66
  67
  68

Sample Output 1:

65-A

66-B

67-C

68-D

Sample Input 2:

- Enter the digits:

  115
  116
  101
  112

Sample Output 2:

115-s

116-t

101-e

112-p

Question-3

Problem Statement – NNRG College wants to recognize the department which has succeeded in getting the maximum number of placements for this academic year. The departments that have participated in the recruitment drive are CSE, AIML, and ECE. Help the college find the department getting maximum placements.

Note: If any input is negative, the output should be "Input is Invalid". If all departments has equal number of placements, the output should be "None of the department has got the highest placement".

Sample Input 1:
- Enter the no of students placed in CSE:90
- Enter the no of students placed in AIML:45
- Enter the no of students placed in ECE:70

Sample Output 1:
- Highest placement
  CSE

Sample Input 2:
- Enter the no of students placed in CSE:55
- Enter the no of students placed in AIML:85
- Enter the no of students placed in ECE:85

Sample Output 2:
- Highest placement
  AIML
  ECE

Sample Input 3:
- Enter the no of students placed in CSE:0
- Enter the no of students placed in AIML:0
- Enter the no of students placed in ECE:0

Sample Output 3:
- None of the department has got the highest placement

Sample Input 4:
- Enter the no of students placed in CSE:10
- Enter the no of students placed in AIML:-50
- Enter the no of students placed in ECE:40

Sample Output 4:
- Input is Invalid

Question-4

Problem Statement – In a Movie theater, there is a discount scheme announced where one gets a 10% discount on the total cost of tickets when there is a bulk booking of more than 20 tickets, and a discount of 2% on the total cost of tickets if a special coupon card is submitted. Develop a program to find the total cost as per the scheme. The cost of the k class ticket is Rs.75 and q class is Rs.150. Refreshments can also be opted by paying an additional of Rs. 50 per member.

Hint: k and q and You have to book minimum of 5 tickets and maximum of 40 at a time. If fails display "Minimum of 5 and Maximum of 40 Tickets". If circle is given a value other than 'k' or 'q' the output should be "Invalid Input".

The ticket cost should be printed exactly to two decimal places.

Sample Input 1:

- Enter the no of ticket:35
- Do you want refreshment:y
- Do you have coupon code:y
- Enter the circle:k

Sample Output 1:

- Ticket cost:4065.25

Sample Input 2:

- Enter the no of ticket:1

Sample Output 2:

- Minimum of 5 and Maximum of 40 Tickets

Question-5

Problem Statement – Sonakshita's teacher has asked her to prepare well for the lesson on seasons. When her teacher tells a month, she needs to say the season corresponding to that month. Write a program to solve the above task.

- Spring – March to May,
- Summer – June to August,
- Autumn – September to November and,
- Winter – December to February.

Month should be in the range 1 to 12. If not the output should be "Invalid month".

Sample Input 1:

- Enter the month:11

Sample Output 1:

- Season:Autumn

Sample Input 2:

- Enter the month:13

Sample Output 2:

- Invalid month

Question-6

Problem Statement – To speed up his composition of generating unpredictable rhythms, Virat Kohli wants the list of prime numbers available in a range of numbers. Can you help him out?

Write a java program to print all prime numbers in the interval [a,b] (a and b, both inclusive).

Note

- Input 1 should be lesser than Input 2. Both the inputs should be positive.
- Range must always be greater than zero.
- If any of the condition mentioned above fails, then display “Provide valid input”
- Use a minimum of one for loop and one while loop

Sample Input 1:

2

15

Sample Output 1:

2 3 5 7 11 13

Sample Input 2:

8

5

Sample Output 2:

- Provide valid input

Question-7

Problem Statement – Virat and Rohit plays by telling numbers. Virat says a number to Rohit. Rohit should first reverse the number and check if it is same as the original. If yes, Rohit should say "Palindrome". If not, he should say "Not a Palindrome". If the number is negative, print "Invalid Input". Help Rohit by writing a program.

Sample Input 1 :

21212

Sample Output 1 :

Palindrome

Sample Input 2 :

6186

Sample Output 2 :

Not a Palindrome

Question-8

NNRG College of Engineering is in the process of increment the salary of the employees. This increment is done based on their salary and their performance appraisal rating.

1. If the appraisal rating is between 1 and 3, the increment is 10% of the salary.
2. If the appraisal rating is between 3.1 and 4, the increment is 25% of the salary.
3. If the appraisal rating is between 4.1 and 5, the increment is 30% of the salary.

Help them to do this, by writing a program that displays the incremented salary. Write a class “IncrementCalculation.java” and write the main method in it.
Note : If either the salary is 0 or negative (or) if the appraisal rating is not in the range 1 to 5 (inclusive), then the output should be “Invalid Input”.

Sample Input 1 :

- Enter the salary

  8000
- Enter the Performance appraisal rating

  3

Sample Output 1 :

8800

Sample Input 2 :

- Enter the salary

  7500
- Enter the Performance appraisal rating

  4.3

Sample Output 2 :

9750

Sample Input 3 :

- Enter the salary

  -5000
- Enter the Performance appraisal rating

  6

Sample Output 3 :

- Invalid Input

Question-9

Problem Statement – Dhoni planned to choose a four digit lucky number for his car. His lucky numbers are 3,5 and 7. Help him find the number, whose sum is divisible by 3 or 5 or 7. Provide a valid car number, Fails to provide a valid input then display that number is not a valid car number.

Note: The input other than 4 digit positive number [includes negative and 0] is considered as invalid.

Refer the samples, to read and display the data.

Sample Input 1:

- Enter the car no:1234

Sample Output 1:

- Lucky Number

Sample Input 2:

- Enter the car no:1214

Sample Output 2:

- Sorry its not my lucky number

Sample Input 3:

- Enter the car no:14

Sample Output 3:

- 14 is not a valid car number

Question-10

Problem Statement –

NNRG institution is offering a variety of courses to students. Students have a facility to check whether a particular course is available in the institution. Write a program to help the institution accomplish this task. If the number is less than or equal to zero display “Invalid Range”.

Assume maximum number of courses is 20.

Sample Input 1:

- Enter no of course:
  5
- Enter course names:
  Java
  Oracle
  C++
  Mysql
  Dotnet
- Enter the course to be searched:
  C++

Sample Output 1:

C++ course is available

Sample Input 2:

- Enter no of course:
  3
- Enter course names:
  Java
  Oracle
  Dotnet
- Enter the course to be searched:
  C++

Sample Output 2:

C++ course is not available

Sample Input 3:

- Enter no of course:
  0

Sample Output 3:

Invalid Range

Question-11

Problem Statement – Sachin Tendulkar buys "N" no of products from a shop. The shop offers a different percentage of discount on each item. He wants to know the item that has the minimum discount offer, so that he can avoid buying that and save money.

[**Input Format**: The first input refers to the no of items; the second input is the item name, price and discount percentage separated by comma(,)]

Assume the minimum discount offer is in the form of Integer.

Note: There can be more than one product with a minimum discount.

Sample Input 1:

4

mobile,10000,20

shoe,5000,10

watch,6000,15

laptop,35000,5

Sample Output 1:

shoe

Explanation: The discount on the mobile is 2000, the discount on the shoe is 500, the discount on the watch is 900 and the discount on the laptop is 1750. So the discount on the shoe is the minimum.

Sample Input 2:

4

Mobile,5000,10

shoe,5000,10

WATCH,5000,10

Laptop,5000,10

Sample Output 2:

Mobile

shoe

WATCH

Laptop

Question-12

Problem Statement – SKY wants to know the maximum marks scored by him in each semester. The mark should be between 0 to 100, if goes beyond the range display "You have entered invalid mark."

Sample Input 1:

- Enter no of semester:
  3
- Enter no of subjects in 1 semester:
  3
- Enter no of subjects in 2 semester:
  2
- Enter no of subjects in 3 semester:
  2
- Marks obtained in semester 1:
  50
  60
  70
- Marks obtained in semester 2:
  90
  98
- Marks obtained in semester 3:
  89
  76

Sample Output 1:

- Maximum mark in 1 semester:70
- Maximum mark in 2 semester:98
- Maximum mark in 3 semester:89

Sample Input 2:

- Enter no of semester:
  3
- Enter no of subjects in 1 semester:
  3
- Enter no of subjects in 2 semester:
  4
- Enter no of subjects in 3 semester:
  2
- Marks obtained in semester 1:
  55
  -56

Sample Output 2:

You have entered invalid mark.

Question-13

Problem Statement – Madhu teaches her daughter to find the factors of a given number. When she provides a number to her daughter, she should tell the factors of that number. Help her to do this, by writing a program. Write a class FindFactor.java and write the main method in it.

Note :

- If the input provided is negative, ignore the sign and provide the output. If the input is zero
- If the input is zero the output should be "No Factors".

Sample Input 1 :

54

Sample Output 1 :

1, 2, 3, 6, 9, 18, 27, 54

Sample Input 2 :

-1869

Sample Output 2 :

1, 3, 7, 21, 89, 267, 623, 1869

Question-14

Problem Statement – An automobile company manufactures both a two wheeler (TW) and a four wheeler (FW). A company manager wants to make the production of both types of vehicle according to the given data below:

- 1st data, Total number of vehicle (two-wheeler + four-wheeler)=v
- 2nd data, Total number of wheels = W

The task is to find how many two-wheelers as well as four-wheelers need to manufacture as per the given data.

Example :

Input :

200 -> Value of V

540 -> Value of W

Output :

TW =130 FW=70

Explanation:
130+70 = 200 vehicles

(70*4)+(130*2)= 540 wheels

Constraints :

- 2<=W
- W%2=0
- V<W

Print “INVALID INPUT” , if inputs did not meet the constraints.

Question-15

Problem Statement – Given a string S(input consisting) of '*' and '#'. The length of the string is variable. The task is to find the minimum number of '*' or '#' to make it a valid string. The string is considered valid if the number of '*' and '#' are equal. The '*' and '#' can be at any position in the string.

Note: The output will be a positive or negative integer based on number of '*' and '#' in the input string.

- (*>#): positive integer
- (#>*): negative integer
- (#=*): 0

Example :

Sample Input 1:

- ###*** -> Value of S

Output :

- 0 → number of * and # are equal

Sample Input 2:

- ###** -> Value of S

Output :

- -1 → number of # are greater than *

Sample Input 3:

- ###***** -> Value of S

Output :

- 1 → number of * are greater than #

Question-16

Problem Statement: Given an integer array Arr of size N the task is to find the count of elements whose value is greater than all of its prior elements.

Note : 1st element of the array should be considered in the count of the result.

For example,

Arr[]={7,4,8,2,9}
As 7 is the first element, it will consider in the result.8 and 9 are also the elements that are greater than all of its previous elements. Since total of 3 elements is present in the array that meets the condition.

Hence the output = 3.

Example 1:

Input
5 -> Value of N, represents size of Arr
7-> Value of Arr[0]
4 -> Value of Arr[1]
8-> Value of Arr[2]
2-> Value of Arr[3]
9-> Value of Arr[4]
Output :
3
Example 2:
5 -> Value of N, represents size of Arr
3 -> Value of Arr[0]
4 -> Value of Arr[1]
5 -> Value of Arr[2]
8 -> Value of Arr[3]
9 -> Value of Arr[4]
Output :
5

Question-17

Problem Statement: A parking lot in a mall has RxC number of parking spaces. Each parking space will either be empty(0) or full(1). The status (0/1) of a parking space is represented as the element of the matrix. The task is to find index of the prpeinzta row(R) in the parking lot that has the most of the parking spaces full(1).

Note :

RxC- Size of the matrix

Elements of the matrix M should be only 0 or 1.

Example 1:

Input :

3 -> Value of R(row)
3 -> value of C(column)
[0 1 0 1 1 0 1 1 1] -> Elements of the array M[R][C] where each element is separated by new line.
Output :
3 -> Row 3 has maximum number of 1's

Example 2:

input :

4 -> Value of R(row)
3 -> Value of C(column)
[0 1 0 1 1 0 1 0 1 1 1 1] -> Elements of the array M[R][C]
Output :
4 -> Row 4 has maximum number of 1's

Question-18

Problem Statement: A party has been organised on cruise. The party is organised for a limited time(T). The number of guests entering (E[i]) and leaving (L[i]) the party at every hour is represented as elements of the array. The task is to find the maximum number of guests present on the cruise at any given instance within T hours.

Example 1:

Input :

- 5 -> Value of T
- [7,0,5,1,3] -> E[], Element of E[0] to E[N-1], where input each element is separated by new line
- [1,2,1,3,4] -> L[], Element of L[0] to L[N-1], while input each element is separate by new line.

Output :

8 -> Maximum number of guests on cruise at an instance.

Explanation:

1st hour:

Entry : 7 Exit: 1

No. of guests on ship : 6

2nd hour :

Entry : 0 Exit : 2

No. of guests on ship : 6-2=4

Hour 3:

Entry: 5 Exit: 1

No. of guests on ship : 4+5-1=8

Hour 4:

Entry : 1 Exit : 3

No. of guests on ship : 8+1-3=6

Hour 5:

Entry : 3 Exit: 4

No. of guests on ship: 6+3-4=5

Hence, the maximum number of guests within 5 hours is 8.

Example 2:

Input:

4 -> Value of T

[3,5,2,0]   -> E[], Element of E[0] to E[N-1], where input each element is separated by new line.

[0,2,4,4]   -> L[], Element of L[0] to L[N-1], while input each element in separated by new line

Output:

6

Cruise at an instance

Explanation:

Hour 1:

Entry: 3 Exit: 0

No. of guests on ship: 3

Hour 2:

Entry : 5 Exit : 2

No. of guest on ship: 3+5-2=6

Hour 3:

Entry : 2 Exit: 4

No. of guests on ship: 6+2-4= 4

Hour 4:

Entry: 0 Exit : 4

No. of guests on ship : 4+0-4=0

Hence, the maximum number of guests within 5 hours is 6.

The input format for testing

The candidate has to write the code to accept 3 input.

First input- Accept value for number of T(Positive integer number)

Second input- Accept T number of values, where each value is separated by a new line.

Third input- Accept T number of values, where each value is separated by a new line.

The output format for testing

The output should be a positive integer number or a message as given in the problem statement(Check the output in Example 1 and Example 2)

Question-19

Problem Statement: A washing machine works on the principle of Fuzzy System, the weight of clothes put inside it for washing is uncertain But based on weight measured by sensors, it decides time and water level which can be changed by menus given on the machine control area.

For low level water, the time estimate is 25 minutes, where approximately weight is between 2000 grams or any nonzero positive number below that.

For medium level water, the time estimate is 35 minutes, where approximately weight is between 2001 grams and 4000 grams.

For high level water, the time estimate is 45 minutes, where approximately weight is above 4000 grams.

Assume the capacity of machine is maximum 7000 grams

Where approximately weight is zero, time estimate is 0 minutes.

Write a function which takes a numeric weight in the range [0,7000] as input and produces estimated time as output is: “OVERLOADED”, and for all other inputs, the output statement is

“INVALID INPUT”.

Input should be in the form of integer value –

Output must have the following format –

Time Estimated: Minutes

Example:
Input value
2000
Output value
Time Estimated: 25 minutes

Question 20

Problem Statement: The Caesar cipher is a type of substitution cipher in which each alphabet in the plaintext or messages is shifted by a number of places down the alphabet.

For example, with a shift of 1, P would be replaced by Q, Q would become R, and so on. To pass an encrypted message from one person to another, it is first necessary that both parties have the 'Key' for the cipher, so that the sender may encrypt and the receiver may decrypt it. Key is the number of OFFSET to shift the cipher alphabet. Key can have basic shifts from 1 to 25 positions as there are 26 total alphabets.

As we are designing custom Caesar Cipher, in addition to alphabets, we are considering numeric digits from 0 to 9. Digits can also be shifted by key places.

For Example, if a given plain text contains any digit with values 5 and key =2, then 5 will be replaced by 7, "-"(minus sign) will remain as it is. Key value less than 0 should result into "INVALID INPUT"

Example 1:

Enter your PlainText: All the best

Enter the Key: 1

The encrypted Text is: Bmm uif Cftu

Question 21

Problem Statement: We want to estimate the cost of painting a property. Interior wall painting cost is Rs.18 per sq.ft. and exterior wall painting cost is Rs.12 per sq.ft.

Take input as
1. Number of Interior walls
2. Number of Exterior walls
3. Surface Area of each Interior 4. Wall in units of square feet
Surface Area of each Exterior Wall in units of square feet

If a user enters zero as the number of walls then skip Surface area values as User may don't want to paint that wall.

Calculate and display the total cost of painting the property
Example 1:
6
3
12.3
15.2
12.3
15.2
12.3
15.2
10.10
10.10
10.00
Total estimated Cost: 1847.4 INR

Question-22

Problem Statement: A doctor has a clinic where he serves his patients. The doctor's consultation fees are different for different groups of patients depending on their age. If the patient's age is below 17, fees is 200 INR. If the patient's age is between 17 and 40, fees is 400 INR. If patient's age is above 40, fees is 300 INR. Write a code to calculate earnings in a day for which one array/List of values representing age of patients visited on that day is passed as input.

Note:

- Age should not be zero or less than zero or above 120
- Doctor consults a maximum of 20 patients a day
- Enter age value (press Enter without a value to stop):

Example 1:
Input
20
30
40
50
2
3
14
Output
Total Income 2000 INR
Note: Input and Output Format should be same as given in the above example.
For any wrong input display INVALID INPUT
Output Format
Total Income 2100 INR

Question-23

Problem Statement:

Explanation:
To check whether a year is leap or not
Step 1:

- We first divide the year by 4.
- If it is not divisible by 4 then it is not a leap year.
- If it is divisible by 4 leaving remainder 0

Step 2:

- We divide the year by 100
- **If it is not divisible by 100 then it is a leap year.**
- If it is divisible by 100 leaving remainder 0

Step 3:

- We divide the year by 400
- If it is not divisible by 400 then it is a leap year.
- If it is divisible by 400 leaving remainder 0

Then it is a leap year

Example 1:

Input
Enter Year: 1896

Output
1896 is a Leap Year

Example 2:

Input
Enter Year: 1900

Output
1900 is not a Leap Year

Question-24

Problem Statement – A chocolate factory is packing chocolates into the packets. The chocolate packets here represent an array of N number of integer values. The task is to find the empty packets(0) of chocolate and push it to the end of the conveyor belt(array).

Example 1 :

N=8 and arr = [4,5,0,1,9,0,5,0].

There are 3 empty packets in the given set. These 3 empty packets represented as O should be pushed towards the end of the array

Input :

8 – Value of N

[4,5,0,1,9,0,5,0] – Element of arr[O] to arr[N-1],While input each element is separated by newline

Output:

4 5 1 9 5 0 0 0

Example 2:

Input:

6 — Value of N.

[6,0,1,8,0,2] – Element of arr[0] to arr[N-1], While input each element is separated by newline

Output:

6 1 8 2 0 0

Question-25

Problem Statement – Virat Kohli is learning digital logic subject which will be for his next semester. He usually tries to solve unit assignment problems before the lecture. Today he got one tricky question. The problem statement is “A positive integer has been given as an input. Convert decimal value to binary representation. Toggle all bits of it after the most significant bit including the most significant bit. Print the positive integer value after toggling all bits”.

Constrains-

1<=N<=100

Example 1:

Input :

10 -> Integer

Output :

5 -> result- Integer

Explanation:

Binary representation of 10 is 1010. After toggling the bits(1010), will get 0101 which represents “5”. Hence output will print “5”.

Question-26

Problem Statement –Rohit Sharma is always excited about sunday. It is favourite day, when he gets to play all day. And goes to cycling with his friends.

So every time when the months starts he counts the number of sundays he will get to enjoy. Considering the month can start with any day, be it Sunday, Monday…. Or so on.

Count the number of Sunday Rohit Sharma will get within n number of days.

Example 1:

Input

mon-> input String denoting the start of the month.

13 -> input integer denoting the number of days from the start of the month.

Output :

2 -> number of days within 13 days.

Explanation:

The month start with mon(Monday). So the upcoming sunday will arrive in next 6 days. And then next Sunday in next 7 days and so on.

Now total number of days are 13. It means 6 days to first sunday and then remaining 7 days will end up in another sunday. Total 2 sundays may fall within 13 days.

Question-27

Problem Statement – A supermarket maintains a pricing format for all its products. A value N is printed on each product. When the scanner reads the value N on the item, the product of all the digits in the value N is the price of the item. The task here is to design the software such that given the code of any item N the product (multiplication) of all the digits of value should be computed(price).

Example 1:

Input :

5244 -> Value of N

Output :

160 -> Price

Explanation:

From the input above

Product of the digits 5,2,4,4

5*2*4*4= 160

Hence, output is 160.

Question-28

Problem Statement – Particulate matters are the biggest contributors to Delhi pollution. The main reason behind the increase in the concentration of PMs includes vehicle emission by applying Odd Even concept for all types of vehicles. The vehicles with the odd last digit in the registration number will be allowed on roads on odd dates and those with even last digit will on even dates.

Given an integer array a[], contains the last digit of the registration number of N vehicles traveling on date D(a positive integer). The task is to calculate the total fine collected by the traffic police department from the vehicles violating the rules.

Note : For violating the rule, vehicles would be fined as X Rs.

Example 1:

Input :

4 -> Value of N

{5,2,3,7} -> a[], Elements a[0] to a[N-1], during input each element is separated by a new line

12 -> Value of D, i.e. date

200 -> Value of x i.e. fine

Output :

600 -> total fine collected

Explanation:

Date D=12 means , only an even number of vehicles are allowed.

Find will be collected from 5,3 and 7 with an amount of 200 each.

Hence, the output = 600.

Question-29

Problem Statement – Write a Program to print the following **“SweetHome.”** Using various methods like roof(), walls() and floor().

```
     *
    * *
   * * *
  * * * *
 * * * * *
#         #
#         #
#         #
#         #
#         #
~~~~~~~~~~~
 SweetHome
```

Question-30

Problem Statement - An intelligence agency has received reports about some threats. The reports consist of numbers in a mysterious method. There is a number “N” and another number “R”. Thosenumbers are studied thoroughly and it is concluded that all digits of the number ‘N’ are summed upand this action is performed ‘R’ number of times. The resultant is also a single digit that is yet to be deciphered. The task here is to find the single-digit sum of the given number ‘N’ by repeating the action ‘R’ number of times.

If the value of ‘R’ is 0, print the output as ‘0’.

Example 1:

Input :

99 -> Value

of N3 ->

Value of R

Output :

9 -> Possible ways to fill the cistern.

Explanation:

Here, the number N=99

1. Sum of the digits N: 9+9 = 18
2. Repeat step 2 ‘R’ times i.e. 3 tims (9+9)+(9+9)+(9+9) = 18+18+18 =54
3. Add digits of 54 as we need a single digit 5+4Hence , the output is 9.

Question-31

Problem Statement - A carry is a digit that is transferred to left if sum of digits exceeds 9 whileadding two numbers from right-to-left one digit at a time

You are required to implement the following

function.int NumberOfCarries(int num1 , int num2);

The functions accepts two numbers ‘num1’ and ‘num2’ as its arguments. You are required to calculate and return the total number of carries generated while adding digits of two numbers ‘num1’and ‘ num2’.

Assumption: num1, num2>=0

Example:

- **Input**
  - Num 1: 451
  - Num 2: 349
- **Output**
  - 2

Explanation:

Adding ‘num 1’ and ‘num 2’ right-to-left results in 2 carries since (1+9) is 10. 1 is carried and(5+4=1) is 10, again 1 is carried. Hence 2 is returned.

Sample Input

Num 1: 23

Num 2: 563

Sample Output

0

Question-32

Problem Statement- Given a 2D array, print it in spiral form. See the following examples.

NOTE:- Please comment down the code in other languages as well below .

Input:

```
1   2   3   4
5   6   7   8
9   10  11  12
13  14  15  16
```

Output:
1 2 3 4 8 12 16 15 14 13 9 5 6 7 11 10

Input:

```
1   2   3   4   5    6
7   8   9  10  11  12
13 14 15 16  17  18
```

Output:
1 2 3 4 5 6 12 18 17 16 15 14 13 7 8 9 10 11

Question-33

Problem Statement- Create a java program that takes in a student's first name and average mark obtained, the program should the group the students using the first vowel in their name.

For example, if the student's name is **Madhu**, then the student belongs to group **A** and if the name was **Uday** then the student belongs to group **U**.

The program should further categorize the mark is follows:

Marks obtained	Grade Level	Message
80 – 100	1	Magnificent!!
70 – 79	2	Excellent!!
60 – 69	3	Good work!!
50 – 59	4	Good!!
0 – 49	0	Fail – Try again next Year!!
Greater than 100 or less than 0	X	Invalid Marks!!, Marks too high. Or Invalid Marks!!, Negative marks not allowed.

Below are samples of the program, for demo purpose only.

Sample run 1:

Enter student full names: **Kiran**

Enter average mark obtained: **88**

Output:

Hi **Kiran**, you were placed in group **I**
Grade Level: **1**

Comment: **Magnificent!!**

Sample run 2:

Enter student full names: **Easwar**

Enter average mark obtained: **200**

Output:

Hi **Easwar**, you were placed in group **E**
Grade Level: **X**

Comment: **Invalid Marks!!, Marks too high.**

Question-34

Problem Statement- Create a Java Program that read Weight (kg) and Height(cm) then display **BMI value** and **Weight Category (Interpretation).** Using **AWT or Swing and Event Handling**.

HOW TO CALCULATE BODY MASS INDEX

$$BMI = \frac{Weight\,(kg)}{[Height(m)]^2}$$

BMI	Interpretation
Below 18.5	Underweight
18.5 - 24.9	Normal
25.0 - 29.9	Overweight
30.0 or greater	Obese

Sample Run 1:

Your Weight(kg) 92

Your Height(cm) 176

Calculate BMI

Your BMI : **29.700413223140497**
This Means : **You are Overweight**

Sample Run 2:

Your Weight(kg) 78

Your Height(cm) 182

Calculate BMI

Your BMI : **23.54788069073783**
This Means : **You are Normal**

Sample Run 3:

Your Weight(kg) 66

Your Height(cm) 192

Calculate BMI

Your BMI : **17.903645833333332**
This Means : **You are Underweight**

Question-35

Problem Statement- Create a Java Program that prompts the user to enter an odd integer value n.The program then should find the smallest even number and the smallest odd number that are divisible by n. Also make sure the square root of obtained numbers previously is greater than 150.

You are required to use the appropriate loop that start from 1 and terminates when both of numbers are found. The Program should display both smallest even and odd numbers.

Below are samples of the program, for demo purpose only.

Sample Run 1:

Enter an odd integer: **97**

The even number is **22504**

The odd number is **22601**

Sample Run 2:

Enter an odd integer: **188**

Enter an odd integer: **25**

The even number is **22550**

The odd number is **22525**

Sample Run 3:

Enter an odd integer: **66**

Enter an odd integer: **75**

The even number is **22650**

The odd number is **22575**

Question-36

Problem Statement- An organization has a system in which the customer IDs consists of 6 digits. At the beginning the first 5 digits are assumed ***d1d2d3d4d5***. The sixth digit, ***d6***, is then calculated using the following *formula:*

$$(\sum di * 2^i) \% 7$$

Where 1 <= i <= 5.

Create a Java Program that prompts *the first 5 digits of Customer ID* and displays the complete 6 – digit Customer ID. Where the read the Customer as an integer.

Below are samples of the program, for demo purpose only.

Sample Run 1:

Enter the first five digits: **56234**

The Customer ID is: **562342**

Sample Run 2:

Enter the first five digits: **12345**

The Customer ID is: **123456**

Sample Run 3:

Enter the first five digits: **23457**

The Customer ID is: **234572**